Structure of Atom

The rich diversity of chemical behaviour of different elements can be traced to the differences in the internal structure of atoms of these elements.

What this unit sets out to do

- know about the discovery of electron, proton and neutron and their characteristics, and describe Thomson, Rutherford and Bohr atomic models
- understand the important features of the quantum mechanical model of atom, the nature of electromagnetic radiation and Planck's quantum theory
- explain the photoelectric effect and describe features of atomic spectra, state the de Broglie relation and Heisenberg uncertainty principle
- define an atomic orbital in terms of quantum numbers, state aufbau principle, Pauli exclusion principle and Hund's rule of maximum multiplicity, and write the electronic configurations of atoms

Where the idea of the atom came from

- the existence of atoms has been proposed since the time of early Indian and Greek philosophers, 400 B.C., who were of the view that atoms are the fundamental building blocks of matter and that the continued subdivisions of matter would ultimately yield atoms which would not be further divisible, the word atom deriving from the Greek a-tomio, meaning uncut-able or non-divisible
- these earlier ideas were mere speculations and there was no way to test them experimentally, so they remained dormant for a very long time and were revived again by scientists in the nineteenth century
- the atomic theory of matter was first proposed on a firm scientific basis by John Dalton, a British school teacher, in 1808, and his theory regarded the atom as the ultimate particle of matter, explaining the law of conservation of mass, the law of constant composition and the law of multiple proportion very successfully
- it failed to explain the results of many experiments, for example that substances like glass or ebonite when rubbed with silk or fur get electrically charged, so this unit starts with the experimental observations made towards the end of the nineteenth and beginning of the twentieth century, which established that atoms are made of sub-atomic particles, electrons, protons and neutrons, a concept very different from that of Dalton

In this unit

2.1 Discovery of Sub-atomic Particles
2.2 Atomic Models
2.3 Developments Leading to Bohr's Model
2.4 Bohr's Model for Hydrogen Atom
2.5 Towards Quantum Mechanical Model
2.6 Quantum Mechanical Model

A sealed glass tube holding two metal electrodes, evacuated to a very low pressure

A high voltage across them starts a current, carried by particles streaming from cathode to anode

Figure 2.A A cathode ray discharge tube

The same tube with a hole cut through the anode, so the stream can pass beyond it

The far end is coated with zinc sulphide, which lights up in a bright spot where the stream lands

Figure 2.B A cathode ray discharge tube with perforated anode

2.1 Discovery of sub-atomic particles

The rule to keep in mind

- an insight into the structure of atom was obtained from the experiments on electrical discharge through gases
- before discussing these results we need to keep in mind a basic rule regarding the behaviour of charged particles, that like charges repel each other and unlike charges attract each other

2.1.1 Discovery of electron

- in 1830 Michael Faraday showed that if electricity is passed through a solution of an electrolyte, chemical reactions occurred at the electrodes which resulted in the liberation and deposition of matter there, and these results suggested the particulate nature of electricity
- in the mid 1850s many scientists began to study electrical discharge in partially evacuated tubes, known as cathode ray discharge tubes, which are made of glass containing two thin pieces of metal called electrodes sealed in it, the discharge being observable only at very low pressures and very high voltages, with the pressure adjusted by evacuation
- when sufficiently high voltage is applied across the electrodes, current starts flowing through a stream of particles moving in the tube from the negative electrode, the cathode, to the positive electrode, the anode, and these were called cathode rays or cathode ray particles
- the flow of current from cathode to anode was further checked by making a hole in the anode and coating the tube behind it with the phosphorescent material zinc sulphide, so that when the rays strike the coating a bright spot is developed

A cathode ray tube with an electric field and a magnetic field set at right angles, both across the electron path

Electric field alone bends the beam to one spot, magnetic field alone to another

Balanced against each other they return the beam to the undeflected spot, and the balance point gives the ratio

Figure 2.C The apparatus to determine the charge to the mass ratio of electron

What the cathode ray experiments showed

- the cathode rays start from cathode and move towards the anode, and they are not themselves visible, but their behaviour can be observed with the help of certain fluorescent or phosphorescent materials which glow when hit by them, television picture tubes being cathode ray tubes whose pictures result from fluorescence on a coated screen
- in the absence of electrical or magnetic field these rays travel in straight lines, and in the presence of such a field their behaviour is similar to that expected from negatively charged particles, suggesting that the cathode rays consist of negatively charged particles, called electrons
- the characteristics of cathode rays do not depend upon the material of electrodes or the nature of the gas present in the tube, so we can conclude that electrons are a basic constituent of all the atoms

2.1.2 Charge to mass ratio of electron

- in 1897 the British physicist J.J. Thomson measured the ratio of electrical charge, e, to the mass of the electron by using a cathode ray tube and applying electrical and magnetic fields perpendicular to each other as well as to the path of the electrons
- the amount of deviation of the particles from their path depends on the magnitude of the negative charge on the particle, greater charge giving greater interaction and greater deflection, on the mass of the particle, the lighter the particle the greater the deflection, and on the strength of the field, the deflection increasing with the voltage across the electrodes or the strength of the magnetic field
- by carrying out accurate measurements of the deflections against field strength, Thomson determined the value of e over me as 1.758820 × 1011 C kg-1, where me is the mass of the electron in kg and e is the magnitude of the charge in coulomb, and since electrons are negatively charged the charge on the electron is -e

2.1.3 Charge on the electron

- R.A. Millikan, 1868-1953, devised a method known as the oil drop experiment, 1906-14, to determine the charge on the electrons
- he found the charge on the electron to be -1.6 × 10-19 C, and the present accepted value of the electrical charge is -1.602176 × 10-19 C
- the mass of the electron was determined by combining this result with Thomson's value of the e over me ratio, giving 9.1094 × 10-31 kg

An atomiser sprays oil as a fine mist into the top of a chamber between two charged plates

X-rays ionise the air inside, and a drop picks up charge by colliding with the gaseous ions

A drop is watched through a telescope while the field is varied, so it can be slowed, hurried or held still

Its charge always comes out as a whole-number multiple of one basic charge

Figure 2.D The Millikan oil drop apparatus for measuring charge 'e'. In chamber, the forces acting on oil drop are: gravitational, electrostatic due to electrical field and a viscous drag force when the oil drop is moving.

2.1.4 Discovery of protons and neutrons

- electrical discharge carried out in the modified cathode ray tube led to the discovery of canal rays carrying positively charged particles, and unlike cathode rays the mass of these particles depends upon the nature of the gas present in the tube, since they are simply the positively charged gaseous ions
- their charge to mass ratio depends on the gas from which they originate, some of them carry a multiple of the fundamental unit of electrical charge, and their behaviour in a magnetic or electrical field is opposite to that observed for the electron
- the smallest and lightest positive ion was obtained from hydrogen and was called the proton, characterised in 1919, and later a need was felt for an electrically neutral particle as one of the constituents of the atom
- these were discovered by Chadwick in 1932 by bombarding a thin sheet of beryllium with α-particles, when electrically neutral particles having a mass slightly greater than that of protons were emitted, and he named these particles neutrons

2.2 Atomic models, and the problems they had to solve

- the observations obtained from these experiments suggested that Dalton's indivisible atom is composed of sub-atomic particles carrying positive and negative charges
- the major problems before the scientists were to account for the stability of the atom, to compare the behaviour of elements in terms of both physical and chemical properties, to explain the formation of different kinds of molecules by the combination of different atoms, and to understand the origin and nature of the characteristics of the electromagnetic radiation absorbed or emitted by atoms

A sphere of positive charge about 10-10 m across, with the charge spread evenly through it

Electrons set into that sphere where the electrostatic arrangement is most stable

Pudding, raisin pudding and watermelon are the names it goes by

Figure 2.E Thomson model of atom

2.2.1 Thomson model of atom

- J.J. Thomson in 1898 proposed that an atom possesses a spherical shape, radius approximately 10-10 m, in which the positive charge is uniformly distributed, with the electrons embedded into it in such a manner as to give the most stable electrostatic arrangement
- an important feature of this model is that the mass of the atom is assumed to be uniformly distributed over the atom, and although it was able to explain the overall neutrality of the atom it was not consistent with the results of later experiments
- Thomson was awarded the Nobel Prize for physics in 1906 for his theoretical and experimental investigations on the conduction of electricity by gases

Different atomic models

- different atomic models were proposed to explain the distributions of these charged particles in an atom, and although some of them were not able to explain the stability of atoms, two are discussed here, one proposed by J.J. Thomson and the other by Ernest Rutherford

Table 2.A Properties of Fundamental Particles

Name	Symbol	Absolute charge/C	Relative charge	Mass/kg	Mass/u	Approx. mass/u
Electron	e	– 1.602176×10–19	–1	9.109382×10–31	0.00054	0
Proton	p	+ 1.602176×10–19	+1	1.6726216×10–27	1.00727	1
Neutron	n	0	0	1.674927×10–27	1.00867	1

A stream of high energy alpha particles from a radioactive source aimed at a very thin gold foil, about 100 nm thick

A circular fluorescent zinc sulphide screen around the foil, flashing wherever a particle lands

Most particles pass straight through, a few are turned aside, and about one in twenty thousand comes almost straight back

Figure 2.F Schematic view of Rutherford's scattering experiment. When a beam of alpha (α) particles is "shot" at a thin gold foil, most of them pass through without much effect. Some, however, are deflected.

The three kinds of ray, and how far they get

- the alpha and beta particles are particles similar to electrons in being particles, while the gamma rays are high energy radiations like X-rays, are neutral in nature and do not consist of particles
- as regards penetrating power, alpha particles are the least, followed by beta rays at a hundred times that of alpha particles, and gamma rays at a thousand times that of alpha particles

2.2.2 Rutherford's nuclear model of atom

What the experiment was, and what it showed

- Rutherford and his students, Hans Geiger and Ernest Marsden, bombarded very thin gold foil with alpha particles from a radioactive source, the foil being surrounded by a circular fluorescent zinc sulphide screen on which each strike made a tiny flash of light
- the results were quite unexpected, because on the Thomson model the mass of each gold atom should have been spread evenly over the whole atom, and the alpha particles had enough energy to pass directly through such a uniform distribution of mass, so it was expected that they would only slow down and change direction by small angles
- what was observed instead was that most of the alpha particles passed through the gold foil undeflected, a small fraction were deflected by small angles, and a very few, about one in twenty thousand, bounced back, that is were deflected by nearly 180°

What Rutherford concluded

- most of the space in the atom is empty, as most of the alpha particles passed through the foil undeflected
- a few positively charged alpha particles were deflected, and that deflection must be due to an enormous repulsive force, showing that the positive charge of the atom is not spread throughout it as Thomson had presumed but is concentrated in a very small volume
- his calculations showed that the volume occupied by the nucleus is negligibly small compared with the total volume of the atom, the radius of the atom being about 10-10 m while that of the nucleus is 10-15 m, a difference you can appreciate by realising that if a cricket ball represents a nucleus then the radius of the atom would be about 5 km

The nuclear model itself

- the positive charge and most of the mass of the atom is densely concentrated in an extremely small region, which Rutherford called the nucleus
- the nucleus is surrounded by electrons that move around it with a very high speed in circular paths called orbits, so the model resembles the solar system, with the nucleus playing the role of the sun and the electrons that of revolving planets
- electrons and the nucleus are held together by electrostatic forces of attraction

2.2.3 Atomic number and mass number, and 2.2.4 isobars and isotopes

Counting the nucleons

- the presence of positive charge on the nucleus is due to the protons in it, the charge on the proton being equal but opposite to that of the electron, and the number of protons in the nucleus is the atomic number, Z, which is 1 for hydrogen and 11 for sodium
- to keep electrical neutrality the number of electrons in an atom is equal to the number of protons, so the atomic number is the number of protons in the nucleus and also the number of electrons in a neutral atom
- while the positive charge of the nucleus is due to protons, the mass of the nucleus is due to protons and neutrons together, which are collectively known as nucleons, and the total number of nucleons is the mass number, A, so that A is the number of protons plus the number of neutrons
- the composition of any atom can be represented by the normal element symbol X with the mass number as a superscript on the left and the atomic number as a subscript on the left

Isobars and isotopes

- isobars are the atoms with the same mass number but different atomic number, for example carbon-14 and nitrogen-14
- atoms with identical atomic number but different mass number are known as isotopes, and the difference between them is due to the presence of a different number of neutrons in the nucleus
- taking hydrogen again, 99.985 per cent of hydrogen atoms contain only one proton and this isotope is called protium, while the rest is made up of deuterium, one proton and one neutron, at 0.015 per cent, and tritium, one proton and two neutrons, which is found in trace amounts on the earth

Two worked problems

- Problem 2.1 asks for the number of protons, neutrons and electrons in bromine-80 with Z = 35, and since the species is neutral the number of protons equals the number of electrons equals 35, while the number of neutrons is 80 minus 35, that is 45
- Problem 2.2 gives a species with 18 electrons, 16 protons and 16 neutrons, so the atomic number is 16 and the element is sulphur, the mass number is 16 plus 16 which is 32, and the species is not neutral because protons and electrons differ, it being an anion with a charge equal to the excess of electrons, 18 minus 16, that is 2
- before using the notation, find out whether the species is a neutral atom, a cation or an anion, and note that the number of neutrons is always the mass number minus the atomic number, whether the species is neutral or an ion

2.2.5 Drawbacks of Rutherford model

Why the planetary picture cannot hold

- the nuclear model is like a small scale solar system, with the nucleus playing the role of the massive sun and the electrons the lighter planets, and the coulomb force between electron and nucleus is mathematically similar to the gravitational force between planets
- however, a body moving in an orbit undergoes acceleration even at constant speed because its direction keeps changing, so an electron in such an orbit is under acceleration, and according to the electromagnetic theory of Maxwell a charged particle under acceleration should emit electromagnetic radiation, a feature that does not arise for planets since they are uncharged
- an electron in an orbit would therefore emit radiation whose energy comes from its own motion, so the orbit would continue to shrink, and calculations show it should take an electron only 10-8 s to spiral into the nucleus, which does not happen, so the Rutherford model cannot explain the stability of an atom

Stationary electrons do not help either

- if the electrons were stationary, the electrostatic attraction between the dense nucleus and the electrons would pull them towards the nucleus, forming a miniature version of Thomson's model of atom
- another serious drawback of the Rutherford model is that it says nothing about the distribution of the electrons around the nucleus, nor about the energies of these electrons

2.3 Developments leading to the Bohr's model of atom

The two developments

- results observed from the studies of interactions of radiations with matter have historically provided immense information regarding the structure of atoms and molecules, and Neils Bohr used these results to improve upon the model proposed by Rutherford
- two developments played a major role, the dual character of the electromagnetic radiation, which means that radiations possess both wave like and particle like properties, and the experimental results regarding atomic spectra

2.3.1 Wave nature of electromagnetic radiation

Maxwell's picture of radiation

- the theory was developed in the early 1870s by James Clerk Maxwell and was experimentally confirmed later by Heinrich Hertz
- Maxwell was the first to give a comprehensive explanation of the interaction between charged bodies and the behaviour of electrical and magnetic fields at the macroscopic level, suggesting that when an electrically charged particle moves under accelaration, alternating electrical and magnetic fields are produced and transmitted, and these fields are transmitted in the form of waves called electromagnetic waves or electromagnetic radiation
- light is the form of radiation known from early days and speculation about its nature dates back to remote ancient times, in earlier days light was supposed by Newton to be made of particles or corpuscules, and it was only in the 19th century that the wave nature of light was established, Maxwell again being the first to reveal that light waves are associated with oscillating electric and magnetic character

The whole run of radiations laid out by wavelength, from radio at one end to gamma at the other

Each named region sits at its own place on that run

Visible light is the narrow strip near 1015 Hz, a small part of the entire spectrum

Figure 2.G (a) The spectrum of electromagnetic radiation. (b) Visible spectrum. The visible region is only a small part of the entire spectrum.

What an electromagnetic wave is like

- the oscillating electric and magnetic fields are perpendicular to each other and both are perpendicular to the direction of propagation of the wave
- unlike sound waves or waves produced in water, electromagnetic waves do not require a medium and can move in vacuum
- there are many types of electromagnetic radiations which differ from one another in wavelength or frequency, and together they constitute what is called the electromagnetic spectrum, whose different regions are identified by different names, the radio frequency region around 106 Hz used for broadcasting, the microwave region around 1010 Hz used for radar, the infrared region around 1013 Hz used for heating, the ultraviolet region around 1016 Hz which is a component of the sun's radiation, and the small portion around 1015 Hz which is ordinarily called visible light, the only part our eyes can see

Frequency, wavelength and the speed of light

- these radiations are characterised by frequency, ν, and wavelength, λ, and the SI unit for frequency is the hertz, Hz or s-1, after Heinrich Hertz, defined as the number of waves that pass a given point in one second
- wavelength has the units of length, and since electromagnetic radiation consists of waves of much smaller wavelengths, smaller units than the metre are used
- in vacuum all types of electromagnetic radiation, regardless of wavelength, travel at the same speed of 3.0 × 108 m s-1, or 2.997925 × 108 m s-1 to be precise, and this is called the speed of light and given the symbol c, the three being related by $c = \nu \lambda$

Wavenumber

- the other commonly used quantity, especially in spectroscopy, is the wavenumber, defined as the number of wavelengths per unit length
- its units are the reciprocal of the wavelength unit, that is m-1, however the commonly used unit is cm-1, which is not an SI unit

Problem 2.3, a radio wavelength

- the Vividh Bharati station of All India Radio, Delhi, broadcasts on a frequency of 1,368 kHz, and the question asks for the wavelength of the radiation emitted and which part of the spectrum it belongs to
- the wavelength is c divided by ν, where c is the speed of electromagnetic radiation in vacuum and ν is the frequency, and substituting the given values gives a characteristic radiowave wavelength

Problems 2.4 and 2.5, from wavelength to frequency

- Problem 2.4 gives the visible spectrum as extending from violet at 400 nm to red at 750 nm and asks for these as frequencies, with 1 nm equal to 10-9 m, and using $c = \nu \lambda$ the frequency of violet light comes to 7.50 × 1014 Hz and that of red light to 4.00 × 1014 Hz, so the range of the visible spectrum is from 4.0 × 1014 to 7.5 × 1014 Hz in frequency units
- Problem 2.5 asks for the wavenumber and the frequency of yellow radiation of wavelength 5800 Å, which is 5800 × 10-8 cm or 5800 × 10-10 m, and the wavenumber follows as the reciprocal of that length while the frequency follows from c divided by the wavelength

2.3.2 Particle nature of electromagnetic radiation, Planck's quantum theory

What the wave picture could not explain

- some experimental phenomena such as diffraction, the bending of a wave around an obstacle, and interference, the combination of two waves to give a wave whose disturbance at each point is the sum of the two, can be explained by the wave nature of the radiation
- these however could not be explained even by the electromagnetic theory of 19th century physics, known as classical physics, the nature of emission of radiation from hot bodies, which is black-body radiation, the ejection of electrons from a metal surface when radiation strikes it, which is the photoelectric effect, and the variation of heat capacity of solids as a function of temperature

Intensity plotted against wavelength, one curve for each temperature

Each curve rises to a maximum and then falls away as the wavelength grows

The hotter the body, the further towards short wavelengths that maximum sits

Figure 2.H Wavelength-intensity relationship

A cavity with one tiny hole and no other opening, the closest practical thing to the ideal

A ray entering the hole is reflected about inside until the walls absorb it

Figure 2.I Black body

Discrete amounts, and hot objects

- these phenomena indicate that the system can take energy only in discrete amounts, and that all possible energies cannot be taken up or radiated, the first concrete explanation for black body radiation having been given by Max Planck in 1900
- hot objects emit electromagnetic radiations over a wide range of wavelengths, and at high temperatures an appreciable proportion of the radiation is in the visible region, a higher proportion of short wavelength blue light being generated as the temperature is raised, which is why an iron rod heated in a furnace first turns dull red, becomes more and more red, then white, and then blue when the temperature becomes very high
- so the intensities of radiations of different wavelengths emitted by a hot body depend upon its temperature, and by the late 1850s it was known that objects made of different material and kept at different temperatures emit different amounts of radiation

What a black body is

- when the surface of an object is irradiated with light, part of the radiant energy is generally reflected as such, part is absorbed and part is transmitted, the reason for incomplete absorption being that ordinary objects are as a rule imperfect absorbers
- an ideal body which emits and absorbs radiations of all frequencies uniformly is called a black body, and the radiation emitted by such a body is called black body radiation, and in practice no such body exists, though carbon black approximates fairly closely to one
- a black body is also a perfect radiator of radiant energy, and it is in thermal equilibrium with its surroundings, radiating the same amount of energy per unit area as it absorbs from its surroundings in any given time
- the amount of light emitted from a black body and its spectral distribution depend only on its temperature, the intensity at a given temperature increasing with wavelength to a maximum and then decreasing, with the maximum shifting to shorter wavelengths as the temperature increases

A clean metal surface inside a vacuum chamber, lit by light of one chosen frequency

Electrons come off the metal and are collected by a detector

The detector counts them and measures the kinetic energy each carries away

Figure 2.J Equipment for studying the photoelectric effect. Light of a particular frequency strikes a clean metal surface inside a vacuum chamber. Electrons are ejected from the metal and are counted by a detector that measures their kinetic energy.

Planck's assumption

- Planck arrived at a satisfactory relationship by assuming that absorption and emmission of radiation arises from oscillators, that is atoms in the wall of the black body, whose frequency of oscillation is changed by interaction with oscillators of electromagnetic radiation
- he assumed that radiation could be sub-divided into discrete chunks of energy, and suggested that atoms and molecules could emit or absorb energy only in discrete quantities and not in a continuous manner, giving the name quantum to the smallest quantity of energy that can be emitted or absorbed in the form of electromagnetic radiation
- the energy E of a quantum of radiation is proportional to its frequency, $E = h\nu$, where the proportionality constant h is known as Planck's constant and has the value $6.626 \times 10\text{-}34$ J s, and with this theory Planck was able to explain the distribution of intensity in black body radiation as a function of frequency or wavelength at different temperatures

The staircase, and the man

- quantisation has been compared to standing on a staircase, where a person can stand on any step but not in between two steps, so the energy can take any one of the values 0, $h\nu$, $2h\nu$, $3h\nu$ and so on up to $nh\nu$, but cannot take on any value between them
- Max Planck, 1858-1947, a German physicist, received his Ph.D in theoretical physics from the University of Munich in 1879, was appointed Director of the Institute of Theoretical Physics at the University of Berlin in 1888, and was awarded the Nobel Prize in Physics in 1918 for his quantum theory, having also made significant contributions in thermodynamics and other areas of physics

The photoelectric effect, and what was seen

- in 1887 H. Hertz performed a very interesting experiment in which electrons were ejected when certain metals, for example potassium, rubidium and caesium, were exposed to a beam of light, and the phenomenon is called the photoelectric effect
- the electrons are ejected from the metal surface as soon as the beam of light strikes it, that is there is no time lag between the striking of the light beam and the ejection of the electrons, and the number of electrons ejected is proportional to the intensity or brightness of the light

Brightness against colour

- for each metal there is a characteristic minimum frequency below which the photoelectric effect is not observed, and while the number of electrons ejected does depend upon the brightness of the light, the kinetic energy of the ejected electrons does not
- red light at 4.3 to 4.6 × 1014 Hz of any brightness may shine on a piece of potassium metal for hours and no photoelectrons are ejected, but as soon as even a very weak yellow light at 5.1 to 5.2 × 1014 Hz shines on it the photoelectric effect is observed, the threshold frequency for potassium being 5.0 × 1014 Hz
- Einstein in 1905 was able to explain the photoelectric effect using Planck's quantum theory of electromagnetic radiation as a starting point

Light as a beam of particles

- shining a beam of light on to a metal surface can be viewed as shooting a beam of particles, the photons, and when a photon of sufficient energy strikes an electron in an atom of the metal it transfers its energy instantaneously to that electron during the collision, so the electron is ejected without any time lag or delay
- the greater the energy possessed by the photon the greater the transfer to the electron and the greater the kinetic energy of the ejected electron, so the kinetic energy of the ejected electron is proportional to the frequency of the radiation
- since the striking photon has energy hν and the minimum energy required to eject the electron is hν0, also called the work function W0, the difference hν minus hν0 is transferred as the kinetic energy of the photoelectron, that is one half me v squared, where me is the mass of the electron and v its velocity
- a more intense beam of light consists of a larger number of photons, so the number of electrons ejected is larger than with a beam of weaker intensity

Dual behaviour, and the man who saw it

- the particle nature of light posed a dilemma, since it could explain black body radiation and the photoelectric effect satisfactorily but was not consistent with the known wave behaviour of light which accounts for interference and diffraction
- the only way to resolve it was to accept that light possesses both particle and wave-like properties, so that depending on the experiment light behaves either as a wave or as a stream of particles, showing particle like properties whenever radiation interacts with matter and wavelike properties when it propagates, and it turns out that some microscopic particles like electrons also exhibit this wave-particle duality
- Albert Einstein, 1879-1955, a German born American physicist, published three research papers in 1905 on special relativity, Brownian motion and the photoelectric effect while employed as a technical assistant in a Swiss patent office in Berne, and received the Nobel Prize in Physics in 1921 for his explanation of the photoelectric effect

Table 2.B Values of Work Function (W0) for a Few Metals

Metal	Li	Na	K	Mg	Cu	Ag
W0 /eV	2.42	2.3	2.25	3.7	4.8	4.3

Problem 2.6, a mole of photons

- the question asks for the energy of one mole of photons of radiation whose frequency is 5 × 1014 Hz
- the energy of one photon is E = hν, with h at 6.626 × 10-34 J s and ν at 5 × 1014 s-1, giving 3.313 × 10-19 J
- the energy of one mole of photons is that multiplied by the Avogadro number, 6.022 × 1023 mol-1, giving 199.51 kJ mol-1

Problem 2.7, photons from a bulb

- a 100 watt bulb emits monochromatic light of wavelength 400 nm, and the question asks for the number of photons emitted per second
- the power of the bulb is 100 J s-1, and the energy of one photon is hν, that is hc divided by λ, which for 400 nm comes to 4.969 × 10-19 J
- dividing the power by the energy per photon gives 2.012 × 1020 photons emitted per second

Problems 2.8 and 2.9, work function and kinetic energy

- in Problem 2.8 radiation of wavelength 300 nm falls on sodium and electrons are emitted with a kinetic energy of 1.68 × 105 J mol-1, and the question asks for the minimum energy needed to remove an electron and the maximum wavelength that will cause emission
- the energy of a 300 nm photon works out to 3.99 × 105 J mol-1, so the minimum energy needed to remove one mole of electrons from sodium is 3.99 minus 1.68, that is 2.31 × 105 J mol-1, and the minimum energy for one electron corresponds through hc over E to a wavelength of 517 nm, which is green light
- in Problem 2.9 the threshold frequency for a metal is 7.0 × 1014 s-1 and the question asks for the kinetic energy of an electron emitted when radiation of frequency 1.0 × 1015 s-1 hits it, and by Einstein's equation the kinetic energy is h times the difference of the two frequencies, that is 6.626 × 10-34 J s times 3.0 × 1014 s-1, giving 1.988 × 10-19 J

2.3.3 Evidence for quantized electronic energy levels, atomic spectra

How a prism spreads white light

- the speed of light depends upon the nature of the medium it passes through, so a beam is deviated or refracted as it passes from one medium to another, and when white light passes through a prism the wave with the shorter wavelength bends more than the one with the longer
- since ordinary white light consists of waves with all the wavelengths in the visible range, a ray of it is spread out into a series of coloured bands called a spectrum, red with the longest wavelength deviated least and violet with the shortest deviated most, and the spectrum we can see runs from violet at 7.50 × 1014 Hz to red at 4 × 1014 Hz
- such a spectrum is called continuous, because violet merges into blue, blue into green and so on, and a similar spectrum is produced when a rainbow forms in the sky
- when radiation interacts with matter, atoms and molecules may absorb energy and reach a higher, unstable energy state, and to return to their normal lower energy state they emit radiations in various regions of the electromagnetic spectrum

Emission and absorption spectra

- the spectrum of radiation emitted by a substance that has absorbed energy is called an emission spectrum, and atoms, molecules or ions that have absorbed radiation are said to be excited
- an absorption spectrum is like the photographic negative of an emission spectrum, a continuum of radiation being passed through a sample which absorbs certain wavelengths, so that the missing wavelengths leave dark spaces in the bright continuous spectrum
- the study of emission or absorption spectra is referred to as spectroscopy, and the emission spectra of atoms in the gas phase do not show a continuous spread from red to violet, rather they emit light only at specific wavelengths with dark spaces between them, which is why such spectra are called line spectra or atomic spectra

A spectrum as a fingerprint, and hydrogen's own

- each element has a unique line emission spectrum, so the characteristic lines can be used in chemical analysis to identify unknown atoms in the same way as fingerprints identify people, and the German chemist Robert Bunsen, 1811-1899, was one of the first to use line spectra to identify elements
- rubidium, caesium, thallium, indium, gallium and scandium were discovered when their minerals were analysed by spectroscopic methods, and helium was discovered in the sun that way
- when an electric discharge is passed through gaseous hydrogen the H2 molecules dissociate and the excited hydrogen atoms produced emit radiation of discrete frequencies, the spectrum consisting of several series of lines named after their discoverers, and Balmer showed in 1885 that the visible lines obey a formula in wavenumber with n an integer equal to or greater than 3

Light from excited hydrogen through a prism, separating into a few discrete wavelengths recorded as bright lines

A sample holds an enormous number of atoms, so between them every possible excited state is represented, and the light they give out falling to lower states makes the spectrum

White light passed through unexcited hydrogen comes out lacking those same wavelengths, a line absorption spectrum that is the negative of the emission one

Figure 2.K (a) Atomic emission. The light emitted by a sample of excited hydrogen atoms (or any other element) can be passed through a prism and separated into certain discrete wavelengths. Thus an emission spectrum, which is a photographic recording of the separated wavelengths is called as line spectrum. Any sample of reasonable size contains an enormous number of atoms. Although a single atom can be in only one excited state at a time, the collection of atoms contains all possible excited states. The light emitted as these atoms fall to lower energy states is responsible for the spectrum. (b) Atomic absorption. When white light is passed through unexcited atomic hydrogen and then through a slit and prism, the transmitted light is lacking in intensity at the same wavelengths as are emitted in (a) The recorded absorption spectrum is also a line spectrum and the photographic negative of the emission spectrum.

From Balmer to Rydberg

- the series of lines described by Balmer's formula are called the Balmer series, and they are the only lines in the hydrogen spectrum which appear in the visible region
- the Swedish spectroscopist Johannes Rydberg noted that all series of lines in the hydrogen spectrum could be described by a single expression, in which n1 runs 1, 2 and so on while n2 takes the values n1 plus 1, n1 plus 2 and so on
- the value 109,677 cm-1 is called the Rydberg constant for hydrogen, and the first five series, for n1 = 1, 2, 3, 4 and 5, are the Lyman, Balmer, Paschen, Bracket and Pfund series
- of all the elements, the hydrogen atom has the simplest line spectrum

Table 2.C The Spectral Lines for Atomic Hydrogen

Series	n1	n2	Spectral Region
Lyman	1	2,3....	Ultraviolet
Balmer	2	3,4....	Visible
Paschen	3	4,5....	Infrared
Brackett	4	5,6....	Infrared
Pfund	5	6,7....	Infrared

A ladder of energy levels for the hydrogen atom, one rung per stationary state

Arrows dropping to n = 1 make the Lyman series, to n = 2 the Balmer, to n = 3 the Paschen

The rungs crowd closer together as n rises

Figure 2.L Transitions of the electron in the hydrogen atom (The diagram shows the Lyman, Balmer and Paschen series of transitions)

What all line spectra share

- the line spectrum becomes more and more complex for heavier atoms, however certain features are common to all line spectra, that the line spectrum of an element is unique, and that there is a regularity in the line spectrum of each element
- the questions that arise are what the reasons for these similarities are and whether it has something to do with the electronic structure of atoms, and the answers provide the key to understanding the electronic structure of these elements

2.4 Bohr's model for hydrogen atom

The first two postulates

- Neils Bohr in 1913 was the first to explain quantitatively the general features of the structure of the hydrogen atom and its spectrum, using Planck's concept of quantisation of energy, and though the theory is not modern quantum mechanics it can still be used to rationalize many points in atomic structure and spectra
- the electron in the hydrogen atom can move around the nucleus in a circular path of fixed radius and energy, and these paths, arranged concentrically around the nucleus, are called orbits, stationary states or allowed energy states
- the energy of an electron in an orbit does not change with time, however the electron will move from a lower stationary state to a higher one when the required amount of energy is absorbed, or emit energy when it moves from a higher stationary state to a lower one, and the energy change does not take place in a continuous manner

Angular momentum, and the last two postulates

- just as linear momentum is the product of mass and linear velocity, angular momentum is the product of moment of inertia and angular velocity, and for an electron of mass me moving in a circular path of radius r the moment of inertia is me r squared and the angular velocity is v over r, so the angular momentum comes to me v r
- the frequency of radiation absorbed or emitted when a transition occurs between two stationary states differing in energy by ΔE follows from that difference divided by h, an expression commonly known as Bohr's frequency rule
- the angular momentum of an electron is quantised, being n times h over 2π with n equal to 1, 2, 3 and so on

Why only certain orbits are allowed

- an electron can move only in those orbits for which its angular momentum is an integral multiple of h over 2π, which is to say the angular momentum is quantised
- radiation is emitted or obsorbed only when a transition of the electron takes place from one quantised value of angular momentum to another, so Maxwell's electromagnetic theory does not apply here, and that is why only certain fixed orbits are allowed

The stationary states, numbered and sized

- the stationary states for the electron are numbered n = 1, 2, 3 and so on, and these integral numbers are known as principal quantum numbers
- the radii of the stationary states are n squared times a0, where a0 is 52.9 pm, so the radius of the first stationary state, called the Bohr orbit, is 52.9 pm, and normally the electron in the hydrogen atom is found in this orbit, that is n = 1, while as n increases the radius increases and the electron is present further from the nucleus

The energy of a stationary state

- the most important property associated with the electron is the energy of its stationary state, given by minus RH divided by n squared, where RH is called the Rydberg constant and has the value 2.18 × 10-18 J
- the energy of the lowest state, also called the ground state, is minus 2.18 × 10-18 J, and the energy of the stationary state for n = 2 is minus 0.545 × 10-18 J
- when the electron is free from the influence of the nucleus the energy is taken as zero, and the electron in this situation is associated with the stationary state of principal quantum number infinity and is called an ionized hydrogen atom

Niels Bohr

- Niels Bohr, 1885-1962, a Danish physicist, received his Ph.D. from the University of Copenhagen in 1911, then spent a year with J.J. Thomson and Ernest Rutherford in England
- in 1913 he returned to Copenhagen where he remained for the rest of his life, and in 1920 he was named Director of the Institute of theoretical Physics
- after the first World War Bohr worked energetically for peaceful uses of atomic energy, received the first Atoms for Peace award in 1957, and was awarded the Nobel Prize in Physics in 1922

Hydrogen-like species

- the negative sign in the energy expression depicts the electron's stability relative to the reference state of zero energy at n = infinity
- Bohr's theory can also be applied to ions containing only one electron, similar to that present in the hydrogen atom, for example He+, Li2+ and Be3+, and the energies of the stationary states of these hydrogen like species carry a factor of Z squared over n squared, while the radii carry 52.9 times n squared over Z pm, where Z is the atomic number, 2 for helium and 3 for lithium
- from these it is evident that the energy becomes more negative and the radius smaller as Z increases, which means the electron will be tightly bound to the nucleus
- it is also possible to calculate the velocities of electrons in these orbits, and qualitatively the magnitude of the velocity increases with increase of positive charge on the nucleus and decreases with increase of principal quantum number

2.4.1 Explanation of line spectrum of hydrogen

Absorbed going up, emitted coming down

- radiation is absorbed if the electron moves from an orbit of smaller principal quantum number to one of higher principal quantum number, whereas radiation is emitted if the electron moves from a higher orbit to a lower one
- the energy gap between the two orbits, ΔE, is the final energy minus the initial energy, and combining that with the expression for the energy of a stationary state gives ΔE as 2.18 × 10-18 J times the difference of one over ni squared and one over nf squared
- the frequency associated with the absorption and emission of the photon follows as 3.29 × 1015 times that same difference in Hz, and in terms of wavenumbers as 1.09677 × 107 times it in m-1

Which way the sign goes

- in the case of an absorption spectrum nf is greater than ni, the term in the parenthesis is positive and energy is absorbed, while in the case of an emission spectrum ni is greater than nf, ΔE is negative and energy is released
- this expression is similar to that used by Rydberg, derived empirically from the experimental data available at that time, and each spectral line, whether in absorption or emission, can be associated with a particular transition in the hydrogen atom
- in a large number of hydrogen atoms different possible transitions can be observed, leading to a large number of spectral lines, and the brightness or intensity of a spectral line depends upon the number of photons of the same wavelength or frequency absorbed or emitted

Problem 2.10, a photon from n = 5 to n = 2

- the question asks for the frequency and wavelength of a photon emitted during a transition from the n = 5 state to the n = 2 state in the hydrogen atom
- since ni is 5 and nf is 2 this transition gives rise to a spectral line in the visible region of the Balmer series, the energy works out to minus 4.58 × 10-19 J, which is an emission energy, and taking the magnitude the frequency of the photon is 6.91 × 1014 Hz

Problem 2.11, the first orbit of He+

- the question asks for the energy associated with the first orbit of He+ and the radius of that orbit
- for He+ the atomic number Z is 2 and n is 1, so the energy comes to minus 8.72 × 10-18 J
- the radius is 0.0529 times n squared over Z nm, which for n = 1 and Z = 2 is 0.02645 nm

2.4.2 Limitations of Bohr's model

What the model could not reach

- Bohr's model of the hydrogen atom was no doubt an improvement over Rutherford's nuclear model, as it could account for the stability and line spectra of the hydrogen atom and of hydrogen like ions such as He+, Li2+ and Be3+, however it was too simple to account for certain points
- it fails to account for the finer details of the hydrogen atom spectrum, the doublet, that is two closely spaced lines, observed by using sophisticated spectroscopic techniques, and it is also unable to explain the spectrum of atoms other than hydrogen, for example the helium atom which possesses only two electrons, while the theory was also unable to explain the splitting of spectral lines in the presence of a magnetic field, the Zeeman effect, or an electric field, the Stark effect
- it could not explain the ability of atoms to form molecules by chemical bonds, so taking these points into account one needs a better theory which can explain the salient features of the structure of complex atoms

2.5 Towards quantum mechanical model of the atom, and 2.5.1 dual behaviour

The two developments, and de Broglie's relation

- in view of the shortcomings of Bohr's model, attempts were made to develop a more suitable and general model for atoms, and two important developments contributed significantly to the formulation of such a model, the dual behaviour of matter and the Heisenberg uncertainty principle
- the French physicist de Broglie in 1924 proposed that matter, like radiation, should also exhibit dual behaviour, that is both particle and wavelike properties, which means that just as the photon has momentum as well as wavelength, electrons should also have momentum as well as wavelength
- from this analogy he gave a relation between the wavelength λ and the momentum p of a material particle, the wavelength being h divided by the momentum

Electron diffraction, and why we never see it in a ball

- it was found that an electron beam undergoes diffraction, a phenomenon characteristic of waves, and this fact has been put to use in making an electron microscope, which is based on the wavelike behaviour of electrons just as an ordinary microscope utilises the wave nature of light, and which is a powerful tool in modern scientific research because it achieves a magnification of about 15 million times
- according to de Broglie every object in motion has a wave character, but the wavelengths associated with ordinary objects are so short, because of their large masses, that their wave properties cannot be detected, whereas the wavelengths associated with electrons and other subatomic particles, with very small mass, can be detected experimentally

Problems 2.12 and 2.13

- Problem 2.12 asks for the wavelength of a ball of mass 0.1 kg moving with a velocity of 10 m s-1, and by the de Broglie equation this is h divided by mass times velocity, giving 6.626 × 10-34 m, which is why no wave behaviour is ever seen in a ball
- Problem 2.13 gives an electron of mass 9.1 × 10-31 kg with a kinetic energy of 3.0 × 10-25 J, and since the kinetic energy is one half m v squared the velocity comes to 812 m s-1, so the wavelength is 8967 × 10-10 m, that is 896.7 nm

Problem 2.14, and the start of uncertainty

- Problem 2.14 asks for the mass of a photon with wavelength 3.6 Å, that is 3.6 × 10-10 m, and taking the velocity of the photon as the velocity of light the mass comes to 6.135 × 10-29 kg
- Werner Heisenberg, a German physicist, in 1927 stated the uncertainty principle, which is the consequence of the dual behaviour of matter and radiation, and it states that it is impossible to determine simultaneously the exact position and the exact momentum, or velocity, of an electron

What the principle says

- the product of the uncertainty in position, Δx, and the uncertainty in momentum or velocity cannot be smaller than a fixed amount set by h
- if the position of the electron is known with a high degree of accuracy, so that Δx is small, then the velocity of the electron will be uncertain, and on the other hand if the velocity is known precisely then the position will be uncertain
- so if we carry out some physical measurement on the electron's position or velocity, the outcome will always depict a fuzzy or blur picture

The metrestick, and the light we would have to use

- the principle can be best understood with an example, that of being asked to measure the thickness of a sheet of paper with an unmarked metrestick, where the results obtained would be extremely inaccurate and meaningless, and to obtain any accuracy you should use an instrument graduated in units smaller than the thickness of the sheet
- analogously, to determine the position of an electron we must use a meterstick calibrated in units smaller than the dimensions of the electron, keeping in mind that an electron is considered a point charge and is therefore dimensionless
- to observe an electron we can illuminate it with light or electromagnetic radiation whose wavelength is smaller than the dimensions of an electron, but the high momentum photons of such light would change the energy of the electron by collision, so we would be able to calculate the position of the electron but would know very little about its velocity after the collision

Significance of the uncertainty principle

- one of the important implications is that it rules out the existence of definite paths or trajectories of electrons and other similar particles
- the trajectory of an object is determined by its location and velocity at various moments, so if we know where a body is at a particular instant and also know its velocity and the forces acting on it at that instant, we can tell where the body would be sometime later, which means the position of an object and its velocity fix its trajectory

Big objects, small objects

- the value of the product of the uncertainties obtained for a heavy object is extremely small and insignificant, so in dealing with milligram-sized or heavier objects the associated uncertainties are hardly of any real consequence
- in the case of a microscopic object like an electron, whose mass is 9.11 × 10-31 kg, the product obtained is much larger and such uncertainties are of real consequence, so that if one tries to find the exact location of the electron to an uncertainty of only 10-8 m, the uncertainty in velocity would be 104 m s-1
- that is so large that the classical picture of electrons moving in fixed Bohr orbits cannot hold good, which means the precise statements of the position and momentum of electrons have to be replaced by statements of the probability that the electron has a given position and momentum, and this is what happens in the quantum mechanical model of atom

Problems 2.15 and 2.16

- Problem 2.15 uses a microscope with suitable photons to locate an electron in an atom within a distance of 0.1 Å and asks for the uncertainty in the measurement of its velocity, which works out to 5.79 × 106 m s-1
- Problem 2.16 gives a golf ball of mass 40 g and speed 45 m per s whose speed can be measured within an accuracy of 2 per cent, and asks for the uncertainty in the position, which comes to 1.46 × 10-33 m, nearly 10 to the 18 times smaller than the diameter of a typical atomic nucleus, so for large particles the uncertainty principle sets no meaningful limit to the precision of measurements

Reasons for the failure of the Bohr model

- in the Bohr model an electron is regarded as a charged particle moving in well defined circular orbits about the nucleus, and the wave character of the electron is not considered
- further, an orbit is a clearly defined path, and this path can be completely defined only if both the position and the velocity of the electron are known exactly at the same time, which is not possible according to the Heisenberg uncertainty principle
- so the Bohr model of the hydrogen atom not only ignores the dual behaviour of matter but also contradicts the Heisenberg uncertainty principle

Erwin Schrödinger

- Erwin Schrödinger, 1887-1961, an Austrian physicist, received his Ph.D. in theoretical physics from the University of Vienna in 1910, and in 1927 succeeded Max Planck at the University of Berlin at Planck's request
- in 1933 he left Berlin because of his opposition to Hitler and Nazi policies and returned to Austria in 1936, and after the invasion of Austria by Germany he was forcibly removed from his professorship, then moved to Dublin, Ireland where he remained for seventeen years
- Schrödinger shared the Nobel Prize for Physics with P.A.M. Dirac in 1933

2.6 Quantum mechanical model of atom

Where classical mechanics stops working

- in view of the inherent weaknesses in the Bohr model there was no point in extending it to other atoms, and what was needed was an insight into the structure of the atom which could account for the wave-particle duality of matter and be consistent with the Heisenberg uncertainty principle, which came with the advent of quantum mechanics
- classical mechanics, based on Newton's laws of motion, successfully describes the motion of all macroscopic objects such as a falling stone or orbiting planets, which have essentially a particle-like behaviour, however it fails when applied to microscopic objects like electrons, atoms and molecules, mainly because it ignores the dual behaviour of matter and the uncertainty principle
- the branch of science that takes this dual behaviour into account is called quantum mechanics, a theoretical science dealing with the motions of microscopic objects that have both observable wave like and particle like properties, and when it is applied to macroscopic objects, for which wave like properties are insignificant, the results are the same as those from classical mechanics

Schrödinger's equation

- quantum mechanics was developed independently in 1926 by Werner Heisenberg and Erwin Schrödinger, and the treatment here is the one based on the ideas of wave motion
- the fundamental equation of quantum mechanics was developed by Schrödinger and won him the Nobel Prize in Physics in 1933, and it incorporates the wave-particle duality of matter as proposed by de Broglie, being quite complex and needing higher mathematics to solve
- for a system such as an atom or a molecule whose energy does not change with time, the equation is written with a mathematical operator called the Hamiltonian, which Schrödinger gave a recipe for constructing from the expression for the total energy of the system, that total energy taking into account the kinetic energies of all the sub-atomic particles, electrons and nuclei, the attractive potential between the electrons and nuclei and the repulsive potential among them

The wave function, and what it is not

- the wave function corresponding to an energy state contains all information about the electron, and it is a mathematical function whose value depends upon the coordinates of the electron in the atom and does not itself carry any physical meaning
- such wave functions of hydrogen or hydrogen like species with one electron are called atomic orbitals, and those pertaining to one-electron species are called one-electron systems, while the probability of finding an electron at a point within an atom is proportional to the square of the modulus of ψ at that point
- the quantum mechanical results for the hydrogen atom successfully predict all aspects of its spectrum, including some phenomena that could not be explained by the Bohr model
- for multi-electron atoms the Schrödinger equation cannot be solved exactly, a difficulty overcome by approximate methods, and such calculations show that orbitals in other atoms do not differ radically from the hydrogen ones, the principal difference being that increased nuclear charge contracts them all somewhat, and that their energies depend on both n and l rather than on n alone

The important features of the model

- the energy of electrons in atoms is quantized, that is it can only have certain specific values, for example when electrons are bound to the nucleus in atoms, and the existence of these quantised electronic energy levels is a direct result of the wave like properties of electrons and are allowed solutions of the Schrödinger wave equation
- both the exact position and exact velocity of an electron in an atom cannot be determined simultaneously, so the path of an electron in an atom can never be determined or known accurately, and one talks only of the probability of finding the electron at different points
- an atomic orbital is the wave function ψ for an electron in an atom, and whenever an electron is described by a wave function we say that the electron occupies that orbital, so since many such wave functions are possible there are many atomic orbitals in an atom, in each of which the electron has a definite energy, and an orbital cannot contain more than two electrons
- in a multi-electron atom the electrons are filled in various orbitals in the order of increasing energy, all the information about an electron being stored in its orbital wave function, and the probability of finding an electron at a point is proportional to the square of the modulus of ψ there, known as the probability density, which is always positive and from which the region where the electron will most probably be found can be predicted

2.6.1 Orbitals and quantum numbers

Size, shape and orientation

- a large number of orbitals are possible in an atom, and qualitatively they can be distinguished by their size, shape and orientation, an orbital of smaller size meaning there is more chance of finding the electron near the nucleus, while shape and orientation mean there is more probability of finding it along certain directions than along others
- atomic orbitals are precisely distinguished by what are known as quantum numbers, and each orbital is designated by three of them, labelled n, l and ml

The principal quantum number

- the principal quantum number n is a positive integer with values 1, 2, 3 and so on, and it determines the size and to a large extent the energy of the orbital, while for the hydrogen atom and hydrogen like species such as He+ and Li2+ the energy and size depend only on n
- it also identifies the shell, and with increase in n the number of allowed orbitals increases and is given by n squared, all the orbitals of a given n constituting a single shell represented by the letters K, L, M and N for n = 1, 2, 3 and 4
- the size of an orbital increases with increase of n, which is to say the electron will be located further from the nucleus, and since energy is required to shift the negatively charged electron away from the positively charged nucleus, the energy of the orbital will increase with n

The azimuthal quantum number

- l is also known as the orbital angular momentum or subsidiary quantum number, and it defines the three-dimensional shape of the orbital
- for a given value of n, l can have n values ranging from 0 to n minus 1, and the number of sub-shells in a principal shell is equal to the value of n, so the first shell has one sub-shell at l = 0, the second has two at l = 0 and 1, the third has three at l = 0, 1 and 2, and so on
- each sub-shell is assigned an azimuthal quantum number, and sub-shells corresponding to l = 0, 1, 2, 3, 4 and 5 are represented by the notations s, p, d, f, g and h

Table 2.D Subshell Notations

n	l	Subshell notation
1	0	1s
2	0	2s
2	1	2p
3	0	3s
3	1	3p
3	2	3d
4	0	4s
4	1	4p
4	2	4d
4	3	4f

How many orbitals a subshell holds

- each orbital in an atom is defined by a set of values for n, l and ml, so an orbital described by n = 2, l = 1, ml = 0 is an orbital in the p sub-shell of the second shell
- the number of orbitals that goes with each subshell is one for s at l = 0, three for p at l = 1, five for d at l = 2, seven for f at l = 3, nine for g at l = 4 and eleven for h at l = 5

Electron spin, the fourth quantum number

- the three quantum numbers labelling an atomic orbital can be used equally well to define its energy, shape and orientation, but they are not enough to explain the line spectra observed in multi-electron atoms, where some lines actually occur in doublets, two closely spaced lines, or triplets, which suggests the presence of a few more energy levels than the three quantum numbers predict
- in 1925 George Uhlenbeck and Samuel Goudsmit proposed a fourth quantum number known as the electron spin quantum number, ms, an electron spinning around its own axis much as the earth spins around its own axis while revolving around the sun, so that besides charge and mass an electron has an intrinsic spin angular quantum number
- the spin angular momentum of the electron is a vector quantity and can have two orientations relative to a chosen axis, distinguished by ms taking the value plus one half or minus one half, these being the two spin states normally represented by two arrows, ↑ for spin up and ↓ for spin down, and two electrons with different ms values are said to have opposite spins, while an orbital cannot hold more than two electrons and those two should have opposite spins

Orbit, orbital, and why the difference matters

- orbit and orbital are not synonymous, an orbit as proposed by Bohr being a circular path around the nucleus in which an electron moves, a precise description of which is impossible according to the Heisenberg uncertainty principle, so Bohr orbits have no real meaning and their existence can never be demonstrated experimentally
- an atomic orbital is a quantum mechanical concept and refers to the one electron wave function ψ in an atom, characterized by the three quantum numbers n, l and ml, its value depending upon the coordinates of the electron, and ψ has by itself no physical meaning
- it is the square of the wave function which has a physical meaning, giving at any point the probability density, that is the probability per unit volume, so that the product of it and a small volume element yields the probability of finding the electron in that volume, the element being kept small because the density varies from one region of space to another
- the total probability of finding the electron in a given volume can then be calculated by the sum of all the products of the density and the corresponding volume elements, and it is thus possible to get the probable distribution of an electron in an orbital

ψ plotted against distance from the nucleus, one curve for 1s and one for 2s

Beside it the same distance axis carrying the probability density instead

The 1s curve falls away steadily from the nucleus, the 2s curve dips, crosses and rises again

Figure 2.M The plots of (a) the orbital wave function ψ(r); (b) the variation of probability density ψ2(r) as a function of distance r of the electron from the nucleus for 1s and 2s orbitals.

The four quantum numbers, summed up

- n defines the shell, determines the size of the orbital and also to a large extent its energy
- there are n subshells in the nth shell, and l identifies the subshell and determines the shape of the orbital, there being 2l plus 1 orbitals of each type in a subshell, that is one s orbital, three p orbitals and five d orbitals per subshell, while to some extent l also determines the energy of the orbital in a multi-electron atom
- ml designates the orientation of the orbital, and for a given l it has 2l plus 1 values, the same as the number of orbitals per subshell, which means the number of orbitals is equal to the number of ways in which they are oriented
- ms refers to the orientation of the spin of the electron

Problem 2.17, counting orbitals in a shell

- the question asks for the total number of orbitals associated with the principal quantum number n = 3
- for n = 3 the possible values of l are 0, 1 and 2, so there is one 3s orbital with ml = 0, three 3p orbitals with ml = -1, 0 and +1, and five 3d orbitals with ml = -2, -1, 0, +1 and +2, giving a total of 1 plus 3 plus 5, that is 9
- the same value can also be obtained from the relation that the number of orbitals is n squared, that is 3 squared equals 9

Problem 2.18, naming orbitals

- the question asks, using s, p, d and f notations, to describe the orbitals with n = 2 and l = 1, n = 4 and l = 0, n = 5 and l = 3, and n = 3 and l = 2
- reading the shell from n and the subshell letter from l, these are the 2p, the 4s, the 5f and the 3d orbitals respectively

2.6.2 Shapes of atomic orbitals

Plotting the wave function against distance

- the orbital wave function ψ for an electron in an atom has no physical meaning, being simply a mathematical function of the coordinates of the electron
- however, for different orbitals the plots of the corresponding wave functions as a function of r, the distance from the nucleus, are different, and the curves for the 1s and 2s orbitals are not alike

Two clouds of dots, one for 1s and one for 2s, thicker where the electron is more likely

Beside them the same two orbitals drawn as smooth closed surfaces

Each surface is a sphere centred on the nucleus, enclosing about 90 per cent of the probability

Figure 2.N (a) Probability density plots of 1s and 2s atomic orbitals. The density of the dots represents the probability density of finding the electron in that region. (b) Boundary surface diagram for 1s and 2s orbitals.

Three identical dumb-bells, each a pair of lobes either side of the nucleus

Axes mutually perpendicular, one lobe pair lying along each of x, y and z

Figure 2.O Boundary surface diagrams of the three 2p orbitals.

What a boundary surface is

- a boundary or contour surface is drawn in space for an orbital on which the value of the probability density is constant, and in principle many such surfaces are possible, but for a given orbital the one taken as a good representation of its shape is the surface enclosing a volume in which the probability of finding the electron is very high, say 90 per cent
- one cannot draw a surface enclosing 100 per cent, because the probability density always has some value, howsoever small, at any finite distance from the nucleus, so no rigid size would do
- the boundary surface for an s orbital is actually a sphere centred on the nucleus, which in two dimensions looks like a circle

s orbitals and p orbitals

- 1s and 2s orbitals are spherical in shape, and in reality all the s-orbitals are spherically symmetric, that is the probability of finding the electron at a given distance is equal in all directions, while the size of the s orbital increases with n, so that 4s is larger than 3s, 3s than 2s and 2s than 1s, the electron being located further from the nucleus as n rises
- for the three 2p orbitals the nucleus is at the origin and, unlike the s-orbitals, the boundary surfaces are not spherical, each p orbital consisting of two sections called lobes on either side of the plane that passes through the nucleus

Four of the five drawn as four-lobed clovers, dxy, dyz, dxz and dx2-y2

The fifth, dz2, different from the rest, a pair of lobes along z with a ring around the middle

All five are equivalent in energy

Figure 2.P Boundary surface diagrams of the five 3d orbitals.

How the three p orbitals differ

- the probability density function is zero on the plane where the two lobes touch each other, and the size, shape and energy of the three orbitals are identical, differing only in the way the lobes are oriented
- since the lobes may be considered to lie along the x, y or z axis they are given the designations 2px, 2py and 2pz, though there is no simple relation between the values of ml and the x, y and z directions, and it is enough to remember that because there are three possible values of ml there are three p orbitals whose axes are mutually perpendicular
- like s orbitals, p orbitals increase in size and energy with the principal quantum number, so 4p is larger and higher than 3p and 3p than 2p, and their probability density functions also pass through zero as the distance from the nucleus increases, the number of radial nodes being n minus 2, that is one for 3p, two for 4p and so on

d orbitals, and counting nodes

- for l = 2 the orbital is known as a d-orbital, and the minimum value of n has to be 3 since l cannot be greater than n minus 1, there being five ml values for l = 2 and thus five d orbitals, designated dxy, dyz, dxz, dx2-y2 and dz2
- the shapes of the first four are similar to each other whereas that of the fifth is different, but all five 3d orbitals are equivalent in energy, and the d orbitals for which n is greater than 3 also have shapes similar to the 3d but differ in energy and size
- besides the radial nodes, the probability density functions for the np and nd orbitals are zero at the planes passing through the nucleus, so for the pz orbital the xy-plane is a nodal plane and for the dxy orbital there are two nodal planes through the origin, these being called angular nodes, whose number is given by l, one for p orbitals, two for d and so on
- the total number of nodes is n minus 1, the sum of l angular nodes and n minus l minus 1 radial nodes

2.6.3 Energies of orbitals

What sets the energy in hydrogen

- the energy of an electron in a hydrogen atom is determined solely by the principal quantum number

Two ladders of levels side by side, hydrogen on the left and a multi-electron atom on the right

On the hydrogen ladder every orbital of the same n sits at the same height

On the other ladder the subshells of one n split apart, s below p below d

Figure 2.Q Energy level diagrams for the few electronic shells of (a) hydrogen atom and (b) multi-electronic atoms. Note that orbitals for the same value of principal quantum number, have the same energies even for different azimuthal quantum number for hydrogen atom. In case of multi-electron atoms, orbitals with same principal quantum number possess different energies for different azimuthal quantum numbers.

Degenerate orbitals in hydrogen

- the energy of the orbitals in the hydrogen atom increases as 1s below 2s and 2p together, below 3s, 3p and 3d together, below 4s, 4p, 4d and 4f together, and so on
- although the shapes of the 2s and 2p orbitals are different, an electron has the same energy in either, and orbitals having the same energy are called degenerate
- the 1s orbital in a hydrogen atom corresponds to the most stable condition and is called the ground state, an electron residing in it being most strongly held by the nucleus, while an electron in the 2s, 2p or higher orbitals is in an excited state

Why multi-electron atoms are different

- the energy of an electron in a multi-electron atom depends not only on its principal quantum number but also on its azimuthal quantum number, so for a given n the s, p, d and f subshells all have different energies, increasing in the order s below p below d below f, and for higher energy levels these differences are pronounced enough that straggering may result, for example 4s below 3d, 6s below 5d and 4f below 6p
- the main reason is the mutual repulsion among the electrons, since the only electrical interaction in hydrogen is the attraction between electron and nucleus, whereas a multi-electron atom also has repulsion terms between every electron and every other, so the stability of an electron there is because the total attractive interactions exceed the repulsive ones
- because of the electrons in the inner shells, an electron in the outer shell does not experience the full positive charge of the nucleus, the effect being lowered by partial screening, which is known as the shielding of the outer shell electrons from the nucleus by the inner shell electrons, and the net positive charge experienced by the outer electrons is known as the effective nuclear charge

Orbitals listed in the order the rule puts them, lowest n plus l first

Where two orbitals share the same n plus l, the one with the smaller n comes first

This is what puts 4s before 3d, and 6s before 5d

Figure 2.R Arrangement of Orbitals with Increasing Energy on the Basis of (n+l) Rule

Shape decides how well an electron shields

- both the attractive and repulsive interactions depend upon the shell and the shape of the orbital the electron is in, so electrons in the spherical s orbital shield the outer electrons more effectively than those in a p orbital, and electrons in p orbitals shield more than those in d orbitals, even though all are in the same shell
- within a shell, because of the spherical shape of the s orbital, an s electron spends more time close to the nucleus than a p electron, which in turn spends more time near it than a d electron, so for a given shell the effective nuclear charge experienced decreases as l increases, the s electron being more tightly bound than the p and the p than the d
- the energy of an electron in an s orbital will therefore be lower, that is more negative, than that of a p electron, which will be lower than that of a d electron, and since the extent of shielding differs between orbitals this leads to the splitting of energy levels within the same shell

The rule, and what it orders

- the dependence of orbital energies on n and l is mathematically quite complicated, but one simple rule is that the lower the value of n plus l for an orbital the lower its energy, and if two orbitals have the same value of n plus l, the one with the lower value of n has the lower energy
- different subshells of a particular shell have different energies in multi-electron atoms, whereas in the hydrogen atom they have the same energy
- energies of the orbitals in the same subshell decrease with increase in atomic number, so the energy of the 2s orbital of hydrogen is greater than that of lithium, that of lithium greater than that of sodium, and so on

2.6.4 Filling of orbitals in atom

Aufbau principle

- the filling of electrons into the orbitals of different atoms takes place according to the aufbau principle, which is based on Pauli's exclusion principle, Hund's rule of maximum multiplicity and the relative energies of the orbitals
- the word aufbau in German means building up, and the building up of orbitals means the filling up of orbitals with electrons

The subshells written out in rows, 1s, then 2s 2p, then 3s 3p 3d, and so on

Diagonal arrows drawn across them from the top right down to the bottom left

Reading along the arrows in turn gives the filling order

Figure 2.S Order of filling of orbitals

What the aufbau principle states

- in the ground state of the atoms, the orbitals are filled in order of their increasing energies, which is to say electrons first occupy the lowest energy orbital available and enter higher energy orbitals only after the lower ones are filled
- the energy of a given orbital depends upon the effective nuclear charge and different types of orbital are affected to different extents, so there is no single ordering of orbital energies universally correct for all atoms, however the order 1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, 5p, 6s, 4f, 5d, 6p, 7s and so on is extremely useful
- with respect to the placement of outermost valence electrons it is remarkably accurate, for example the valence electron in potassium must choose between 3d and 4s and is found in 4s as the sequence predicts, though the order should be assumed a rough guide, since in many cases orbitals are similar in energy and small changes in atomic structure may change the order of filling, so exceptions may occur

Pauli exclusion principle

- the number of electrons to be filled in various orbitals is restricted by the exclusion principle, given by the Austrian scientist Wolfgang Pauli in 1926, according to which no two electrons in an atom can have the same set of four quantum numbers
- it can also be stated as only two electrons may exist in the same orbital and these electrons must have opposite spin, which means the two can share the same n, l and ml but must have the opposite spin quantum number
- this restriction lets us calculate the capacity of any subshell, so the 1s subshell comprises one orbital and holds at most two electrons, while p and d subshells hold at most 6 and 10, and it sums up as the maximum number of electrons in the shell with principal quantum number n being 2n squared

Hund's rule of maximum multiplicity

- this rule deals with the filling of electrons into the orbitals belonging to the same subshell, that is orbitals of equal energy, called degenerate orbitals
- it states that pairing of electrons in the orbitals belonging to the same subshell does not take place until each orbital belonging to that subshell has got one electron each, that is is singly occupied

Where pairing starts

- since there are three p, five d and seven f orbitals, the pairing of electrons will start in the p, d and f orbitals with the entry of the 4th, 6th and 8th electron respectively
- it has been observed that half filled and fully filled degenerate sets of orbitals acquire extra stability due to their symmetry

2.6.5 Electronic configuration of atoms

Two ways of writing it down

- the distribution of electrons into the orbitals of an atom is called its electronic configuration, and if one keeps in mind the basic rules governing the filling of orbitals, the configurations of different atoms can be written very easily
- in the first notation the subshell is represented by its letter symbol with the number of electrons present written as a superscript, and the same subshell in different shells is distinguished by writing the principal quantum number before it
- in the second notation, the orbital diagram, each orbital of the subshell is represented by a box and the electron by an arrow, ↑ for a positive spin or ↓ for a negative one, and the advantage of this over the first is that it represents all four quantum numbers

Building up from hydrogen to argon

- the hydrogen atom has only one electron, which goes into the orbital with the lowest energy, so its configuration is 1s1, and the second electron in helium can also occupy the 1s orbital, giving 1s2, the two differing from each other in having opposite spin
- the third electron of lithium is not allowed in the 1s orbital because of the Pauli exclusion principle, so it takes the next available choice, the 2s orbital, giving 1s2 2s1, and since 2s can accommodate one more electron the configuration of beryllium is 1s2 2s2
- in the next six elements, boron, carbon, nitrogen, oxygen, fluorine and neon, the 2p orbitals get progressively filled, a process completed with the neon atom at 1s2 2s2 2p6
- the elements from sodium at 1s2 2s2 2p6 3s1 to argon at 1s2 2s2 2p6 3s2 3p6 follow exactly the same pattern as lithium to neon, with the difference that the 3s and 3p orbitals are getting filled now, and this can be simplified by representing the total number of electrons in the first two shells by the name of the element neon

Core and valence electrons, and the transition series

- sodium to argon can be written as neon in brackets followed by the 3s and 3p electrons, the electrons in the completely filled shells being known as core electrons and those added to the shell with the highest principal quantum number as valence electrons
- in potassium and calcium the 4s orbital, being lower in energy than the 3d orbitals, is occupied by one and two electrons respectively, and a new pattern begins with scandium, where the 3d orbital being lower in energy than 4p is filled first, so in the next ten elements from scandium through titanium, vanadium, chromium, manganese, iron, cobalt, nickel and copper to zinc the five 3d orbitals are progressively occupied
- chromium and copper have five and ten electrons in 3d rather than four and nine as their position would indicate, because fully filled and half-filled orbitals have extra stability, that is lower energy, so configurations such as p3, p6, d5, d10, f7 and f14 are more stable, and chromium and copper therefore adopt the d5 and d10 configurations, though exceptions do exist

Onward to the heaviest elements

- with the saturation of the 3d orbitals the filling of the 4p orbital starts at gallium and is complete at krypton, and in the next eighteen elements from rubidium to xenon the pattern of filling 5s, 4d and 5p is similar to that of 4s, 3d and 4p
- then comes the turn of the 6s orbital, holding one electron in caesium and two in barium, and from lanthanum to mercury the filling takes place in 4f and 5d orbitals, after which come 6p, then 7s and finally 5f and 6d, while the elements after uranium are all short-lived and all of them are produced artificially

2.6.6 Stability of completely filled and half filled subshells

Why knowing the configuration matters, and where it bends

- the modern approach to chemistry depends almost entirely on electronic distribution to understand and explain chemical behaviour, so questions such as why atoms combine into molecules, why some elements are metals and others non-metals, and why helium and argon are unreactive while the halogens are reactive find simple explanations in the electronic configuration, none of which the Daltonian model could answer
- the ground state electronic configuration of an atom always corresponds to the state of the lowest total electronic energy, and most atoms follow the basic rules, however in certain elements such as Cu or Cr, where the 4s and 3d subshells differ slightly in energy, an electron shifts from the lower subshell to the higher one provided the shift leaves the higher subshell either completely filled or half filled
- the valence electronic configurations of Cr and Cu are therefore 3d5 4s1 and 3d10 4s1 rather than 3d4 4s2 and 3d9 4s2, and there is extra stability associated with these configurations

Five degenerate orbitals each holding one electron, all spins parallel

Every pair of those electrons can swap places, and each swap releases a little energy

The count of possible swaps is largest exactly when the subshell is half filled or full

Figure 2.T Possible exchange for a d5 configuration

Symmetrical distribution of electrons

- it is well known that symmetry leads to stability, and the completely filled or half filled subshells have a symmetrical distribution of electrons in them and are therefore more stable
- electrons in the same subshell have equal energy but different spatial distribution, so their shielding of one another is relatively small and the electrons are more strongly attracted by the nucleus

Exchange energy

- the stabilizing effect arises whenever two or more electrons with the same spin are present in the degenerate orbitals of a subshell, since these electrons tend to exchange their positions and the energy released due to this exchange is called exchange energy
- the number of exchanges that can take place is maximum when the subshell is either half filled or completely filled, so the exchange energy is maximum and so is the stability
- the exchange energy is at the basis of Hund's rule, that electrons which enter orbitals of equal energy have parallel spins

Every element in order of atomic number, with the configuration each one settles into

Read across a row: symbol, atomic number, and the subshells filled

Elements whose configuration departs from the aufbau order are marked with an asterisk

Figure 2.U Electronic Configurations of the Elements

What this page carries

- the electronic configurations of the known elements, as determined by spectroscopic methods, are tabulated here and continue on the next page
- the asterisk in the table marks elements with exceptional electronic configurations, the ones where an electron shifts to leave a subshell half filled or completely filled
- the table is printed as a drawing rather than as text, so the configurations themselves are in the book and not repeated here

The table, continued

- the second half of the table of electronic configurations runs on from the first, again in order of atomic number
- elements with atomic number 112 and above have been reported but not yet fully authenticated and named, and the table marks them with a double asterisk

Summary

From Dalton to Rutherford

- atoms are the building blocks of elements, the smallest parts of an element that chemically react, and the first atomic theory, proposed by John Dalton in 1808, regarded the atom as the ultimate indivisible particle of matter
- towards the end of the nineteenth century it was proved experimentally that atoms are divisible and consist of three fundamental particles, electrons, protons and neutrons, and the discovery of these sub-atomic particles led to the proposal of various atomic models
- Thomson in 1898 proposed that an atom consists of a uniform sphere of positive electricity with electrons embedded into it, a model in which the mass is considered evenly spread over the atom, and it was proved wrong by Rutherford's alpha-particle scattering experiment in 1909, from which he concluded that the atom is made of a tiny positively charged nucleus at its centre with electrons revolving around it in circular orbits
- the Rutherford model, which resembles the solar system, was an improvement over Thomson's but could not account for the stability of the atom, that is why the electron does not fall into the nucleus, and it was also silent about the distribution and relative energies of electrons around the nucleus

What resolved those difficulties

- the difficulties of the Rutherford model were overcome by Niels Bohr, whose postulates rest on Planck's quantum theory, on the dual nature of electromagnetic radiation, and on the experimental results of atomic spectra
- the quantum mechanical model that followed treats the electron by a wave function, gives up definite paths in favour of probability, and describes each orbital by the quantum numbers n, l, ml and ms
- filling those orbitals by the aufbau principle, the Pauli exclusion principle and Hund's rule of maximum multiplicity gives the electronic configuration of every element, which is what the modern explanation of chemical behaviour rests on

Summary, concluded

- orbitals are filled in order of increasing energy in accordance with the Pauli exclusion principle, that no two electrons in an atom can have the same set of four quantum numbers, and Hund's rule of maximum multiplicity, that pairing of electrons in the orbitals belonging to the same subshell does not take place until each orbital of that subshell has got one electron each, that is is singly occupied
- this forms the basis of the electronic structure of atoms

Exercises

Exercises 2.1 to 2.5

- 2.1 (i) Calculate the number of electrons which will together weigh one gram. (ii) Calculate the mass and charge of one mole of electrons.
- 2.2 (i) Calculate the total number of electrons present in one mole of methane. (ii) Find (a) the total number and (b) the total mass of neutrons in 7 mg of 14C. (Assume that mass of a neutron = 1.675 × 10–27 kg). (iii) Find (a) the total number and (b) the total mass of protons in 34 mg of NH3 at STP. Will the answer change if the temperature and pressure are changed ?
- 2.3 How many neutrons and protons are there in the following nuclei ? 13C, 16O, 24Mg, 56Fe, 88Sr
- 2.4 Write the complete symbol for the atom with the given atomic number (Z) and atomic mass (A) (i) Z = 17, A = 35. (ii) Z = 92, A = 233. (iii) Z = 4, A = 9.
- 2.5 Yellow light emitted from a sodium lamp has a wavelength (λ) of 580 nm. Calculate the frequency (ν) and wavenumber of the yellow light.

Exercises 2.6 to 2.11

- 2.6 Find energy of each of the photons which (i) correspond to light of frequency 3×1015 Hz. (ii) have wavelength of 0.50 Å.
- 2.7 Calculate the wavelength, frequency and wavenumber of a light wave whose period is 2.0 × 10–10 s.
- 2.8 What is the number of photons of light with a wavelength of 4000 pm that provide 1J of energy?
- 2.9 A photon of wavelength 4 × 10–7 m strikes on metal surface, the work function of the metal being 2.13 eV. Calculate (i) the energy of the photon (eV), (ii) the kinetic energy of the emission, and (iii) the velocity of the photoelectron (1 eV= 1.6020 × 10–19 J).
- 2.10 Electromagnetic radiation of wavelength 242 nm is just sufficient to ionise the sodium atom. Calculate the ionisation energy of sodium in kJ mol–1.
- 2.11 A 25 watt bulb emits monochromatic yellow light of wavelength of 0.57μm. Calculate the rate of emission of quanta per second.
- 2.12 Electrons are emitted with zero velocity from a metal surface when it is exposed to radiation of wavelength 6800 Å. Calculate threshold frequency (ν0) and work function (W0) of the metal.
- 2.13 What is the wavelength of light emitted when the electron in a hydrogen atom undergoes transition from an energy level with n = 4 to an energy level with n = 2?

Exercises, continued

Exercises 2.14 to 2.19

- 2.14 How much energy is required to ionise a H atom if the electron occupies n = 5 orbit? Compare your answer with the ionization enthalpy of H atom (energy required to remove the electron from n =1 orbit).
- 2.15 What is the maximum number of emission lines when the excited electron of a H atom in n = 6 drops to the ground state?
- 2.16 (i) The energy associated with the first orbit in the hydrogen atom is –2.18 × 10–18 J atom–1. What is the energy associated with the fifth orbit? (ii) Calculate the radius of Bohr's fifth orbit for hydrogen atom.
- 2.17 Calculate the wavenumber for the longest wavelength transition in the Balmer series of atomic hydrogen.
- 2.18 What is the energy in joules, required to shift the electron of the hydrogen atom from the first Bohr orbit to the fifth Bohr orbit and what is the wavelength of the light emitted when the electron returns to the ground state? The ground state electron energy is –2.18 × 10–11 ergs.
- 2.19 The electron energy in hydrogen atom is given by En = (–2.18 × 10–18)/n2 J. Calculate the energy required to remove an electron completely from the n = 2 orbit. What is the longest wavelength of light in cm that can be used to cause this transition?

Exercises 2.20 to 2.26

- 2.20 Calculate the wavelength of an electron moving with a velocity of 2.05 × 107 m s–1.
- 2.21 The mass of an electron is 9.1 × 10–31 kg. If its K.E. is 3.0 × 10–25 J, calculate its wavelength.
- 2.22 Which of the following are isoelectronic species i.e., those having the same number of electrons? Na+, K+, Mg2+, Ca2+, S2–, Ar.
- 2.23 (i) Write the electronic configurations of the following ions: (a) H– (b) Na+ (c) O2– (d) F– (ii) What are the atomic numbers of elements whose outermost electrons are represented by (a) 3s1 (b) 2p3 and (c) 3p5 ? (iii) Which atoms are indicated by the following configurations ? (a) [He] 2s1 (b) [Ne] 3s2 3p3 (c) [Ar] 4s2 3d1.
- 2.24 What is the lowest value of n that allows g orbitals to exist?
- 2.25 An electron is in one of the 3d orbitals. Give the possible values of n, l and ml for this electron.
- 2.26 An atom of an element contains 29 electrons and 35 neutrons. Deduce (i) the number of protons and (ii) the electronic configuration of the element.

Exercises 2.27 to 2.30

- 2.27 Give the number of electrons in the species listed.
- 2.28 (i) An atomic orbital has n = 3. What are the possible values of l and ml ? (ii) List the quantum numbers (ml and l) of electrons for 3d orbital. (iii) Which of the following orbitals are possible? 1p, 2s, 2p and 3f
- 2.29 Using s, p, d notations, describe the orbital with the following quantum numbers. (a) n=1, l=0; (b) n = 3; l=1 (c) n = 4; l =2; (d) n=4; l=3.
- 2.30 Explain, giving reasons, which of the following sets of quantum numbers are not possible. (a) n = 0, l = 0, ml = 0, ms = + ½ (b) n = 1, l = 0, ml = 0, ms = – ½ (c) n = 1, l = 1, ml = 0, ms = + ½ (d) n = 2, l = 1, ml = 0, ms = – ½

Exercises, continued

Exercises 2.30 continued, and 2.31 to 2.37

- 2.30, continued: (e) n = 3, l = 3, ml = –3, ms = + ½ (f) n = 3, l = 1, ml = 0, ms = + ½
- 2.31 How many electrons in an atom may have the following quantum numbers? (a) n = 4, ms = – ½ (b) n = 3, l = 0
- 2.32 Show that the circumference of the Bohr orbit for the hydrogen atom is an integral multiple of the de Broglie wavelength associated with the electron revolving around the orbit.
- 2.33 What transition in the hydrogen spectrum would have the same wavelength as the Balmer transition n = 4 to n = 2 of He+ spectrum ?
- 2.34 Calculate the energy required for the process He+ (g) giving He2+ (g) + e–. The ionization energy for the H atom in the ground state is 2.18 × 10–18 J atom–1
- 2.35 If the diameter of a carbon atom is 0.15 nm, calculate the number of carbon atoms which can be placed side by side in a straight line across length of scale of length 20 cm long.
- 2.36 2 ×108 atoms of carbon are arranged side by side. Calculate the radius of carbon atom if the length of this arrangement is 2.4 cm.
- 2.37 The diameter of zinc atom is 2.6 Å. Calculate (a) radius of zinc atom in pm and (b) number of atoms present in a length of 1.6 cm if the zinc atoms are arranged side by side lengthwise.

Exercises 2.38 to 2.47

- 2.38 A certain particle carries 2.5 × 10–16C of static electric charge. Calculate the number of electrons present in it.
- 2.39 In Milikan's experiment, static electric charge on the oil drops has been obtained by shining X-rays. If the static electric charge on the oil drop is –1.282 × 10–18C, calculate the number of electrons present on it.
- 2.40 In Rutherford's experiment, generally the thin foil of heavy atoms, like gold, platinum etc. have been used to be bombarded by the α-particles. If the thin foil of light atoms like aluminium etc. is used, what difference would be observed from the above results ?
- 2.41 Symbols with the mass number above and the atomic number below can be written, whereas the reverse arrangement is not acceptable. Answer briefly.
- 2.42 An element with mass number 81 contains 31.7% more neutrons as compared to protons. Assign the atomic symbol.
- 2.43 An ion with mass number 37 possesses one unit of negative charge. If the ion conatins 11.1% more neutrons than the electrons, find the symbol of the ion.
- 2.44 An ion with mass number 56 contains 3 units of positive charge and 30.4% more neutrons than electrons. Assign the symbol to this ion.
- 2.45 Arrange the following type of radiations in increasing order of frequency: (a) radiation from microwave oven (b) amber light from traffic signal (c) radiation from FM radio (d) cosmic rays from outer space and (e) X-rays.
- 2.46 Nitrogen laser produces a radiation at a wavelength of 337.1 nm. If the number of photons emitted is 5.6 × 1024, calculate the power of this laser.
- 2.47 Neon gas is generally used in the sign boards. If it emits strongly at 616 nm, calculate (a) the frequency of emission, (b) distance traveled by this radiation in 30 s (c) energy of quantum and (d) number of quanta present if it produces 2 J of energy.

Exercises, continued

Exercises 2.48 to 2.54

- 2.48 In astronomical observations, signals observed from the distant stars are generally weak. If the photon detector receives a total of 3.15 × 10–18 J from the radiations of 600 nm, calculate the number of photons received by the detector.
- 2.49 Lifetimes of the molecules in the excited states are often measured by using pulsed radiation source of duration nearly in the nano second range. If the radiation source has the duration of 2 ns and the number of photons emitted during the pulse source is 2.5 × 1015, calculate the energy of the source.
- 2.50 The longest wavelength doublet absorption transition is observed at 589 and 589.6 nm. Calcualte the frequency of each transition and energy difference between two excited states.
- 2.51 The work function for caesium atom is 1.9 eV. Calculate (a) the threshold wavelength and (b) the threshold frequency of the radiation. If the caesium element is irradiated with a wavelength 500 nm, calculate the kinetic energy and the velocity of the ejected photoelectron.
- 2.52 Following results are observed when sodium metal is irradiated with different wavelengths. Calculate (a) threshold wavelength and, (b) Planck's constant. λ (nm) 500, 450, 400 against v × 10–5 (cm s–1) 2.55, 4.35, 5.35
- 2.53 The ejection of the photoelectron from the silver metal in the photoelectric effect experiment can be stopped by applying the voltage of 0.35 V when the radiation 256.7 nm is used. Calculate the work function for silver metal.
- 2.54 If the photon of the wavelength 150 pm strikes an atom and one of tis inner bound electrons is ejected out with a velocity of 1.5 × 107 m s–1, calculate the energy with which it is bound to the nucleus.

Exercises 2.55 to 2.62

- 2.55 Emission transitions in the Paschen series end at orbit n = 3 and start from orbit n and can be represeted as v = 3.29 × 1015 (Hz) [1/32 – 1/n2]. Calculate the value of n if the transition is observed at 1285 nm. Find the region of the spectrum.
- 2.56 Calculate the wavelength for the emission transition if it starts from the orbit having radius 1.3225 nm and ends at 211.6 pm. Name the series to which this transition belongs and the region of the spectrum.
- 2.57 Dual behaviour of matter proposed by de Broglie led to the discovery of electron microscope often used for the highly magnified images of biological molecules and other type of material. If the velocity of the electron in this microscope is 1.6 × 106 ms–1, calculate de Broglie wavelength associated with this electron.
- 2.58 Similar to electron diffraction, neutron diffraction microscope is also used for the determination of the structure of molecules. If the wavelength used here is 800 pm, calculate the characteristic velocity associated with the neutron.
- 2.59 If the velocity of the electron in Bohr's first orbit is 2.19 × 106 ms–1, calculate the de Broglie wavelength associated with it.
- 2.60 The velocity associated with a proton moving in a potential difference of 1000 V is 4.37 × 105 ms–1. If the hockey ball of mass 0.1 kg is moving with this velocity, calcualte the wavelength associated with this velocity.
- 2.61 If the position of the electron is measured within an accuracy of + 0.002 nm, calculate the uncertainty in the momentum of the electron. Suppose the momentum of the electron is h/4πm × 0.05 nm, is there any problem in defining this value.
- 2.62 The quantum numbers of six electrons are given below. Arrange them in order of increasing energies. If any of these combination(s) has/have the same energy lists: 1. n = 4, l = 2, ml = –2 , ms = –1/2 2. n = 3, l = 2, ml = 1 , ms = +1/2

Exercises, concluded

Exercise 2.62, continued

- 3. n = 4, l = 1, ml = 0 , ms = +1/2
- 4. n = 3, l = 2, ml = –2 , ms = –1/2
- 5. n = 3, l = 1, ml = –1 , ms = +1/2
- 6. n = 4, l = 1, ml = 0 , ms = +1/2

Exercises 2.63 to 2.67

- 2.63 The bromine atom possesses 35 electrons. It contains 6 electrons in 2p orbital, 6 electrons in 3p orbital and 5 electron in 4p orbital. Which of these electron experiences the lowest effective nuclear charge ?
- 2.64 Among the following pairs of orbitals which orbital will experience the larger effective nuclear charge? (i) 2s and 3s, (ii) 4d and 4f, (iii) 3d and 3p.
- 2.65 The unpaired electrons in Al and Si are present in 3p orbital. Which electrons will experience more effective nuclear charge from the nucleus ?
- 2.66 Indicate the number of unpaired electrons in : (a) P, (b) Si, (c) Cr, (d) Fe and (e) Kr.
- 2.67 (a) How many subshells are associated with n = 4 ? (b) How many electrons will be present in the subshells having ms value of –1/2 for n = 4 ?